

CMEP Project: Training a Conditional Variational Autoencoder on 2D Ising Model Data

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1 Summary

The machine learning model developed is a Conditional Variational Autoencoder (CVAE) that uses CNN layers for image like data. The objective was to train the model to capture the phase transition dynamics of the Ising model and generate new datasets with physically realistic observables. The CVAE model was built using the Keras 3 library with Tensorflow 2.7 for the backend and data pipeline.

1.1 Data Pipeline

The training dataset was generated by a C code that implements a simulation of the 2D Ising model on a 32x32 lattice using the Wolff cluster algorithm. It outputs two files, one containing the binary (integer) lattice samples and the other the relative inverse temperatures, or beta values. The lattice indices and beta indices are split for allocating the training/validation datasets using stratified sampling and the training/validation datasets are loaded into RAM with a buffer and batched. Data augmentation and float casting for the integer spins are applied on-the-fly to each batch during training.

1.2 Training and Hyperparameter Tuning

Training uses the standard Evidence Lower Bound (ELBO) objective with a binary cross-entropy (BCE) reconstruction loss for the input and output lattices and a Kullback-Leibler (KL) divergence between the prior and learned posterior gaussians. Training uses a train/validation split of 0.9, normalized beta values, batch size of 128, 100 epochs and Adam optimizer. Hyperparameter tuning with Optuna was applied to the KL loss weight and the learning rate by minimizing the unweighted validation loss for 50 trials.

1.3 Model Architecture

Components	Details
Encoder Inputs	Image: (32, 32, 1), β condition: (1,1)
Early Embedding	Tile β to (32, 32, 1) $\rightarrow$ Concatenate with image channel for (32, 32, 2) tensor
Encoder CNN	Conv2D layers (filters: [32, 64, 128], kernel: 3x3, strides: 2)
Encoder MLP	Flatten $\rightarrow$ Hidden dense (256) $\rightarrow$ Dense (32) $\rightarrow \mu, \log(\sigma^2)$ layers
Latent Space	$z \in \mathbb{R}^{32}$, Custom sampling layer (reparameterization trick)
Encoder Outputs	$\mu, \log(\sigma^2), z$
Decoder Inputs	Latent code z : (32, 1), β condition: (1, 1)
Late Embedding	Concatenate z and β to (33, 1) tensor
Decoder MLP	Dense (256) $\rightarrow$ Dense (2048), Reshape to Encoder CNN tail shape
Decoder CNN	Conv2DTranspose (filters: [128, 64, 32], kernel: 3x3, strides: 2), FiLM $\rightarrow$ flatten
Decoder Output	(32, 32, 1) logit values

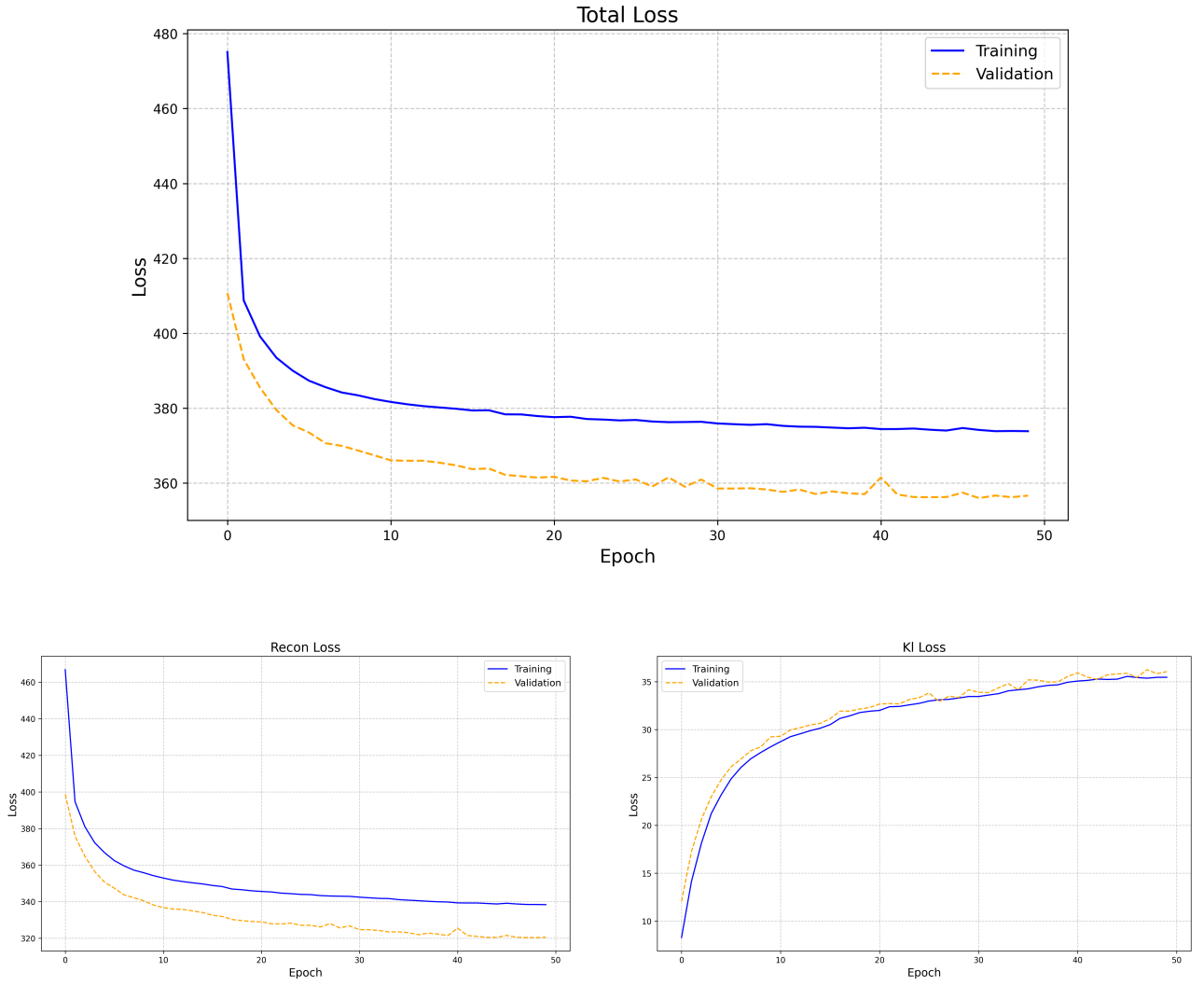
Table 1: CVAE Architecture Summary.

2 Results

2.1 Hyperparameter Tuning Values

Parameter	Search Space	Best Value
Learning Rate	$[10^{-6}, 10^{-3}]$	5×10^{-5}
KL Weight (α)	$[0.1, 5.0]$	1.0

2.2 Training Dynamics



2.3 Observables

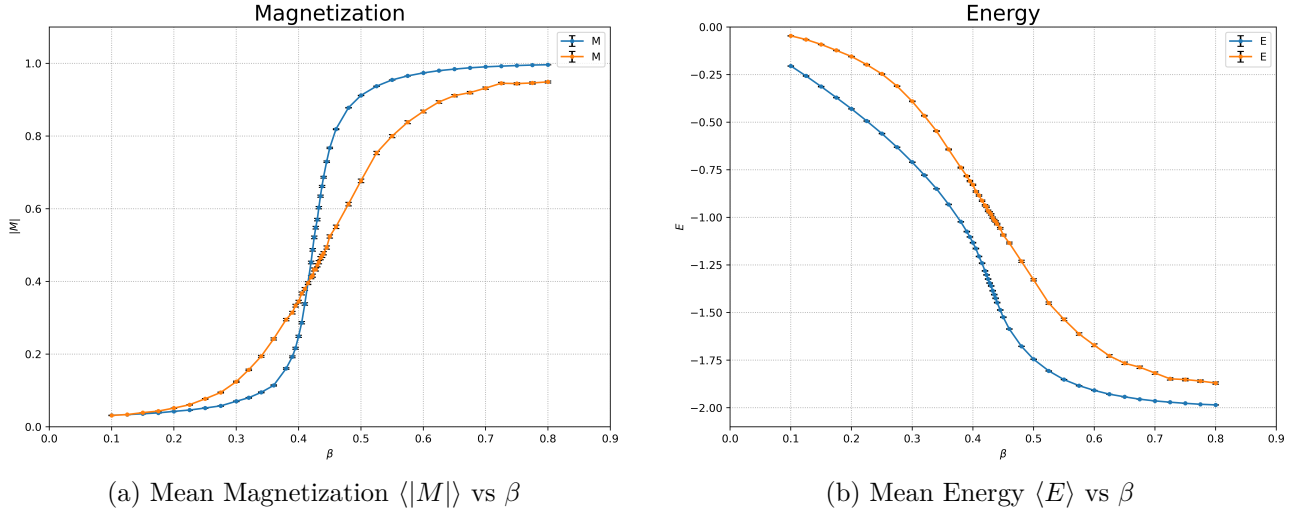


Figure 2: Comparison of CVAE and Ising observables

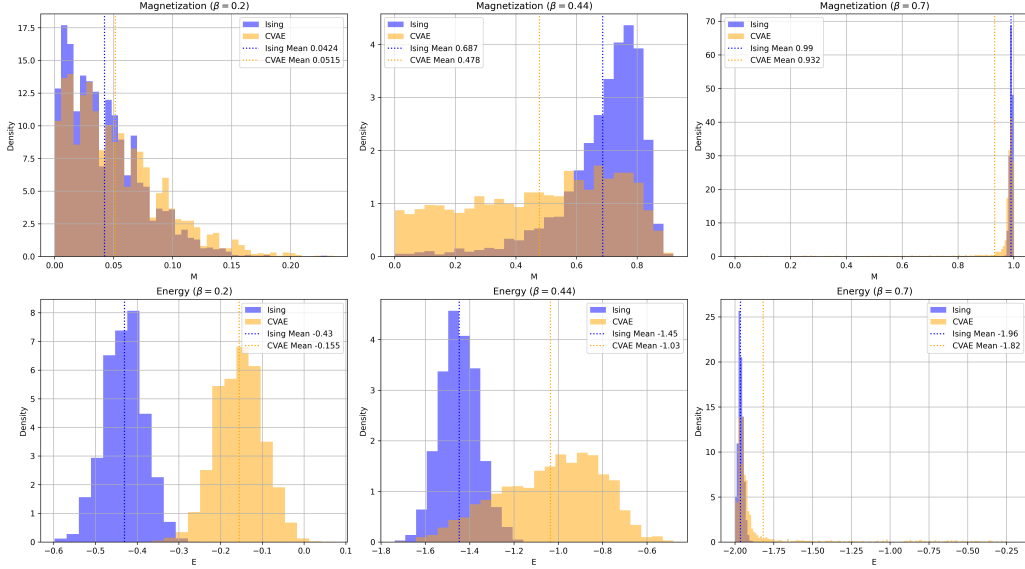


Figure 3: Energy (Top) and Magnetization (Bottom) histograms for selected temperatures: Disordered ($\beta = 0.2$), Critical ($\beta = 0.43$), and Ordered ($\beta = 0.8$). The CVAE struggles to capture the magnetization and energy distributions at low temperatures in a similar way, showing long tails that corresponds to unphysical lattice configurations. At higher temperatures the energy is off by a positive offset, showing the model's inability to sharply capture local correlations.

3 Reconstructions

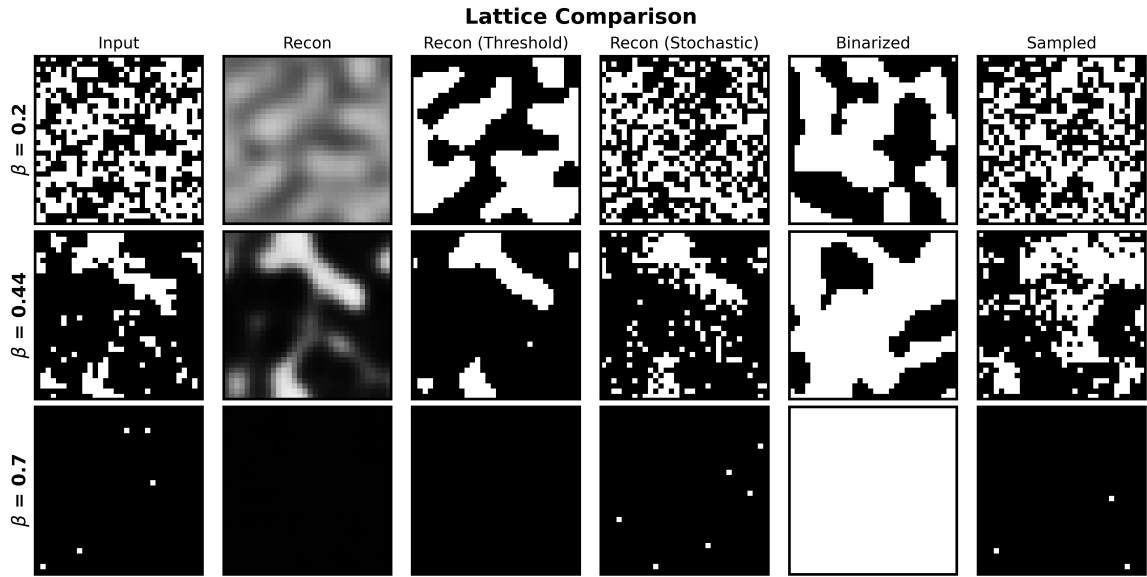


Figure 4: Input vs. Reconstructions for different phases: Disordered ($\beta = 0.2$), Critical ($\beta = 0.43$), and Ordered ($\beta = 0.8$). The reconstructions are for the float CVAE output, binarized output, stochastically sampled binary output, and the CVAE generated binarized and stochastic outputs.

